

Recurrence Relation

A Recurrence relation for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence;

Example: 1 $S = \{5, 8, 11, 14, 17, \dots\}$ is
 $a_n = a_{n-1} + 3$, $n \geq 2$ with initial condition $a_1 = 5$

Find the first four terms of each of the following recurrence relation.

(i) $a_r = 2a_{r-1} + k$, for all integers
 $r \geq 2, a_1 = 1$

(ii) $a_r = 2a_{r-1} + 3a_{r-2}, \forall k \geq 2, a_0 = 1, a_1 = 2$

Order of the Recurrence Relation: The order of a Linear recurrence relation is the difference between the largest and the smallest subscript appearing in the relation.

Example: $a_n - 3a_{n-1} + 2a_{n-2} = 0$ is a recurrence relation of order two.

Example: $(a_n)^2 + (a_{n-1})^2 = 1$ order is 1

Degree of the Recurrence Relation: The degree of a recurrence relation is the highest power of a_n .

Example: $(a_n)^2 + (a_{n-1})^2 = 1$ degree is 2

Example: $a_n - 3a_{n-1} + 2a_{n-2} = 0$ is a recurrence relation of degree one.

Linear: A recurrence relation is called linear if its degree is one.

In general, a recurrence relation of the form $c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} = f(n)$ is linear where $c_0, c_1, c_2, \dots, c_k$ are constant.

Homogeneous Recurrence Relation: the above recurrence relation is said to be homogeneous of order k if $f(n) = 0$

Non homogeneous Recurrence Relation: the above recurrence relation is said to be non-homogeneous of order k if $f(n) \neq 0$

Solution of linear recurrence relation with constant coefficients: $c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} = f(n)$

The solution is $a = a_n^h + a_n^p$
 a_n^h can be find by letting $f(n) = 0$ and the characteristic is
$$c_0 \alpha^k + c_1 \alpha^{k-1} + \dots + c_k = 0$$

case-1: Distinct roots

$$a_n^h = A_1 \alpha_1^n + A_2 \alpha_2^n + \dots + A_k \alpha_k^n$$

case-11: Repeated roots

$$a_n^h = (A_1 + A_2 n + A_3 n^2 + \dots) \alpha^n$$

case-111: complex roots

$$a_n^h = A_1 (a + ib)^n + A_2 (a - ib)^n$$

case-1: Distinct roots

$$a_n^h = A_1 \alpha_1^n + A_2 \alpha_2^n + \dots + A_k \alpha_k^n$$

Example: Solve $a_n - 6a_{n-1} + 8a_{n-2} = 0$

Example: $a_n - 5a_{n-1} + 6a_{n-2} = 0$, $a_0 = 2, a_1 = 5$

case-11: Repeated roots

$$a_n^h = (A_1 + A_2n + A_3n^2 + \dots)\alpha^n$$

Example: Solve $a_n - 6a_{n-1} + 9a_{n-2} = 0$

Solve the Fibonacci sequence

$a_n = a_{n-1} + a_{n-2}$ $n \geq 2$ with initial conditions $a_0 = 1, a_1 = 1$

Particular Solution: Case-1: $f(n)$ is a polynomial of degree r

$$f(n) = a + bn + cn^2 + dn^3 + \dots + en^r$$

then the particular solution assumed as

$$a_n^p = A + Bn + Cn^2 + Dn^3 + \dots + En^r$$

Find the solution of the recurrence relation

$$a_n + 5a_{n-1} + 6a_{n-2} = 3n^2$$

Solution: $a_n + 5a_{n-1} + 6a_{n-2} = 3n^2$
the solution of above recurrence relation is
 $a_n = a_n^h + a_n^p$

a_n^h can be find by assuming $a_n + 5a_{n-1} + 6a_{n-2} = 0$
the characteristic equation of given recurrence relation is
 $\alpha^2 + 5\alpha + 6 = 0 \Rightarrow \alpha = -2, -3$
 $\therefore a_n^h = A_1(-2)^n + A_2(-3)^n$

The a_n^p of $a_n + 5a_{n-1} + 6a_{n-2} = 3n^2$ can be found by assuming $a_n^p = A + Bn + Cn^2$

$$\Rightarrow a_{n-1}^p = A + B(n-1) + C(n-1)^2$$

$$\Rightarrow a_{n-2}^p = A + B(n-2) + C(n-2)^2$$

$$\begin{aligned} a_{n-1}^p &= A + B(n-1) + C(n-1)^2 \\ &= A + nB - B + Cn^2 + C - 2nC \\ &= Cn^2 + (B - 2C)n + A - B + C \end{aligned}$$

$$\begin{aligned}
 \Rightarrow a_{n-2}^p &= A + B(n-2) + C(n-2)^2 \\
 &= A + nB - 2B + Cn^2 + 4C - 4Cn \\
 &= Cn^2 + (B - 4C)n + (A + 4C - 2B)
 \end{aligned}$$

$$\begin{aligned}
 \therefore a_n + 5a_{n-1} + 6a_{n-2} &= 3n^2 \text{ is} \\
 A + Bn + Cn^2 + 5(Cn^2 + (B - 2C)n + A - B + C) &+ 6(Cn^2 + (B
 \end{aligned}$$

$$\Rightarrow A + Bn + Cn^2 + 5Cn^2 + 5Bn - 10Cn + 5A - 5B + 5C + 6Cn^2 + 6Bn - 24Cn + 6A + 24C - 12B = 3n^2$$

$$\Rightarrow (A + 5A - 5B + 5C + 6A + 24C - 12B) + (B + 5B - 10C + 6B - 24C)n + (c + 6c)$$

$$\begin{aligned} \Rightarrow & (A + 5A - 5B + 5C + 6A + 24C - 12B) \\ & + (B + 5B - 10C + 6B - 24C)n \\ & + (c + 6c + 5c)n^2 = 3n^2 \end{aligned}$$

$$\begin{aligned} \Rightarrow & (12A - 17B + 29C) + (12B - 34C)n \\ & + (12c)n^2 = 3n^2 \end{aligned}$$

$$\Rightarrow (12A - 17B + 29C) = 0$$

$$\Rightarrow 12B - 34C = 0$$

$$\Rightarrow 12C = 3$$

$$\Rightarrow C = \frac{1}{4}$$

$$\Rightarrow B = \frac{34}{48} = \frac{17}{24}$$

$$\Rightarrow A = \frac{115}{288}$$

$$\therefore a^p_n = \frac{115}{288} + \frac{17}{24}n + \frac{1}{4}n^2$$

$$\therefore a_n = a_n^h + a_n^p$$

$$a_n = A_1(-2)^n + A_2(-3)^n + \frac{115}{288} + \frac{17}{24}n + \frac{1}{4}n^2$$

Find the particular solution of the recurrence relation

$$a_n - 5a_{n-1} + 6a_{n-2} = 1$$

Find the particular solution of the recurrence relation

$$a_n - a_{n-1} - 2a_{n-2} = n^2$$